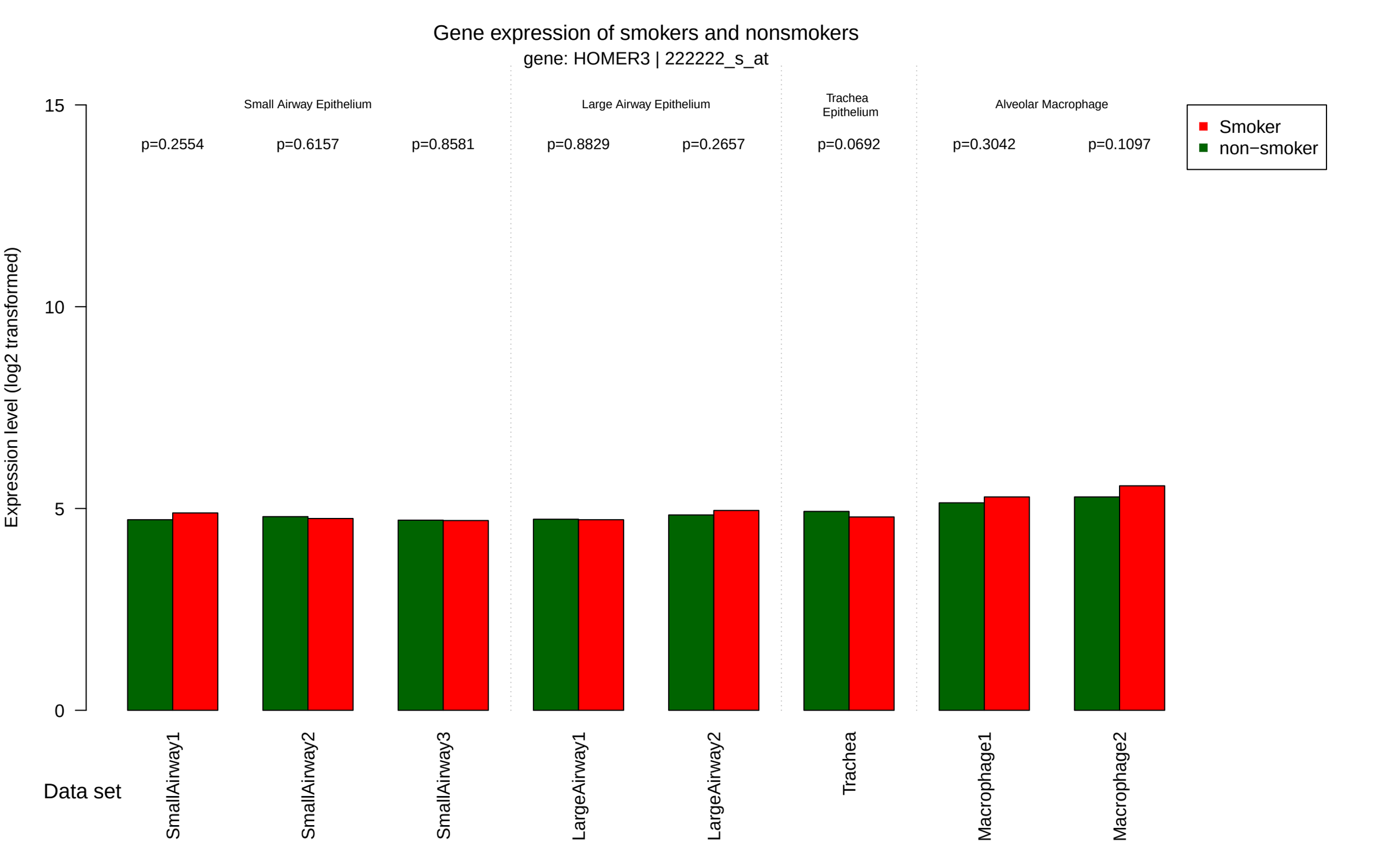
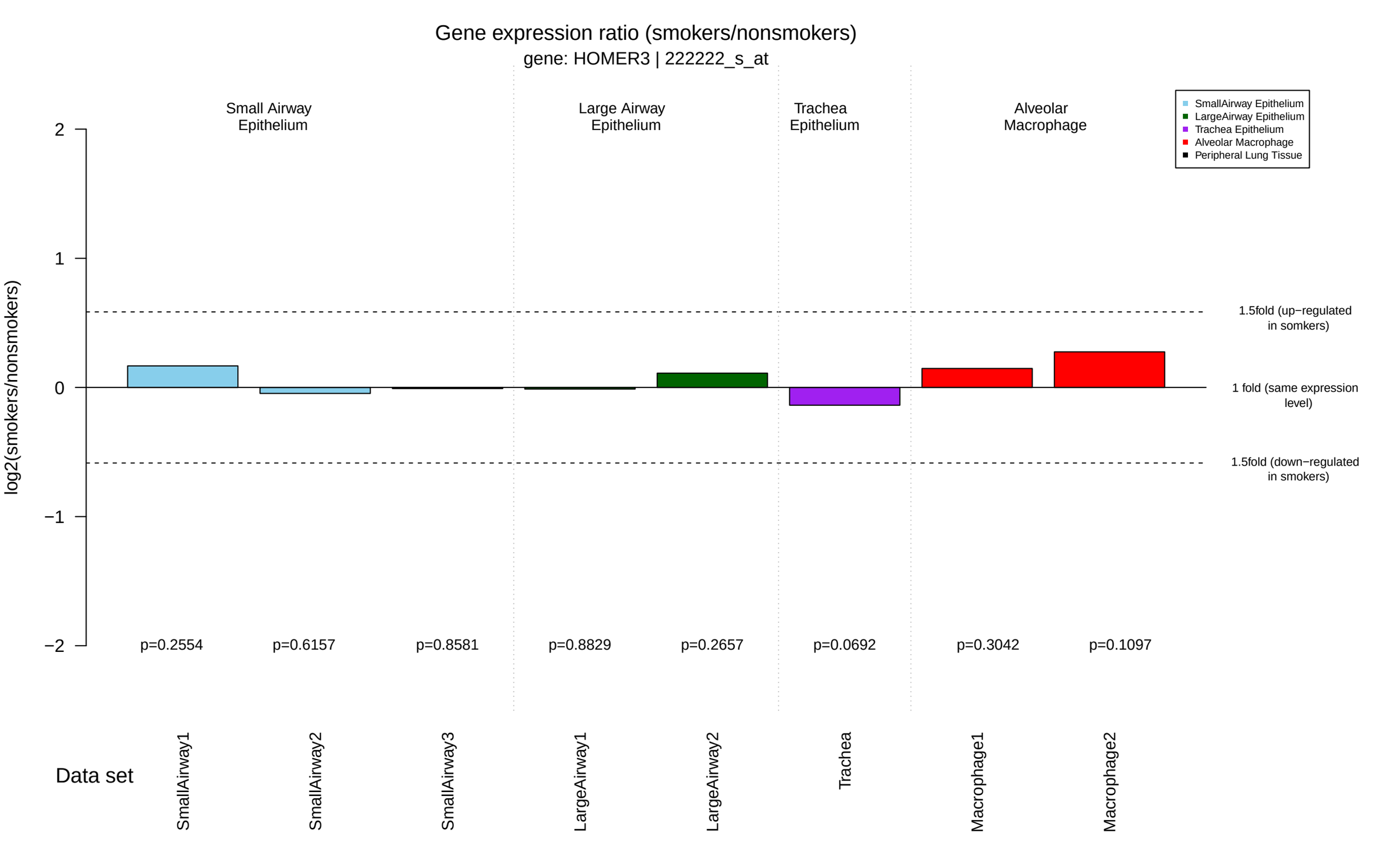
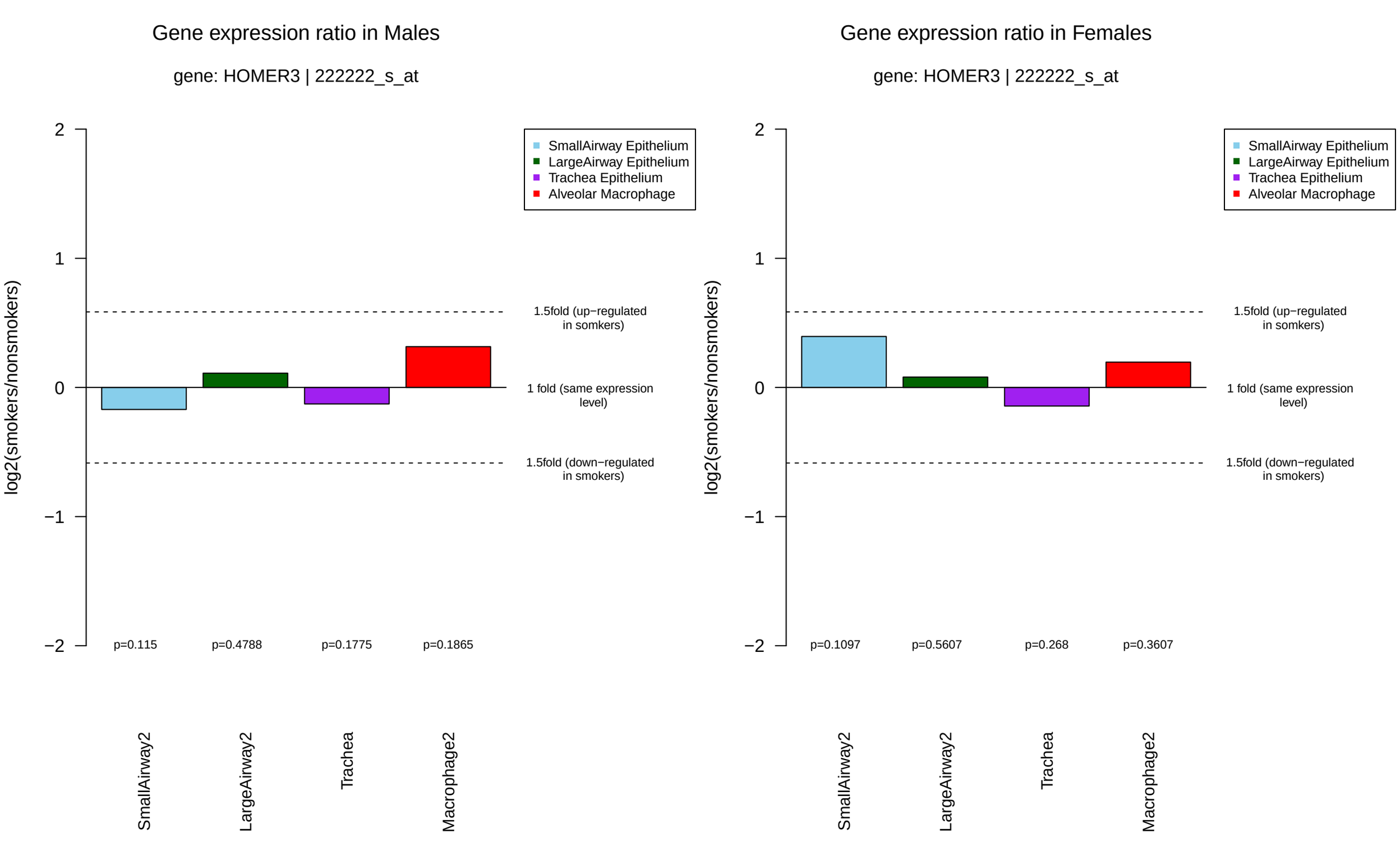
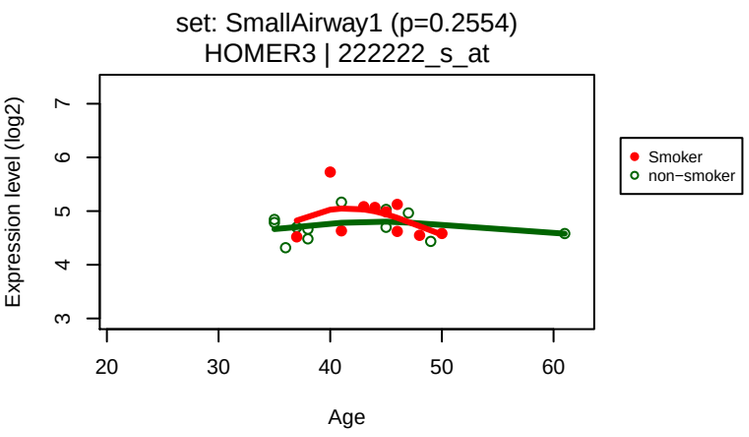
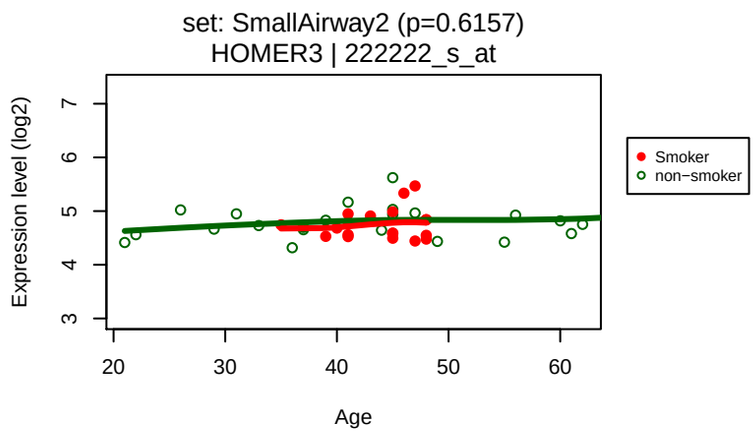
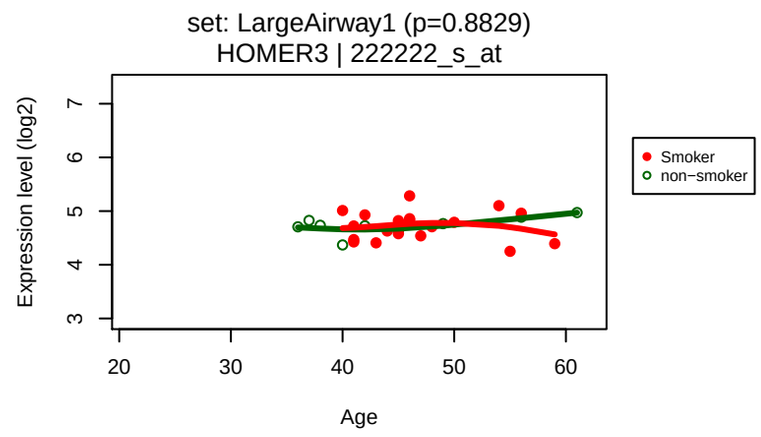

Gene expression of smokers and nonsmokers
gene: HOMER3 | 222222_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2554
p=0.6157
p=0.8581
p=0.8829
p=0.2657
p=0.0692
p=0.3042
p=0.1097
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

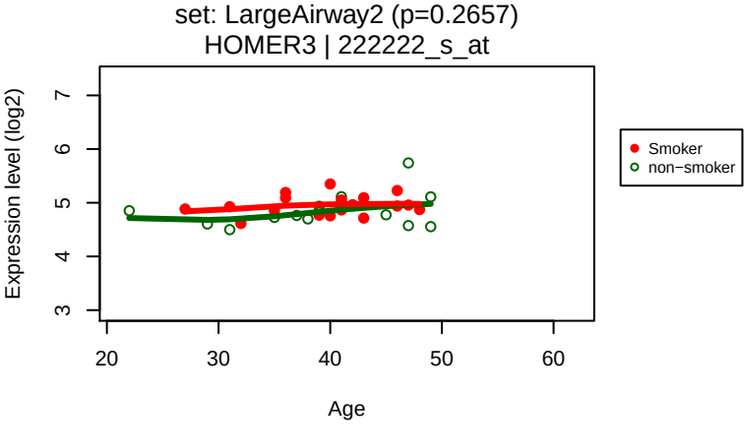

Gene expression ratio (smokers/nonsmokers)
gene: HOMER3 | 222222_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2554
p=0.6157
p=0.8581
p=0.8829
p=0.2657
p=0.0692
p=0.3042
p=0.1097
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

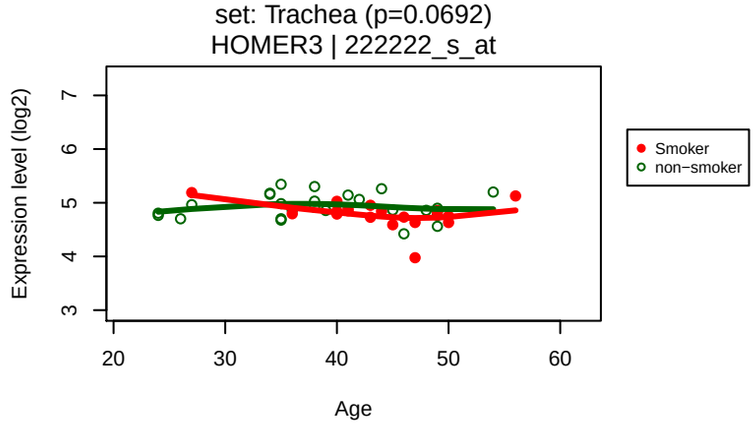

Gene expression ratio in Males
gene: HOMER3 | 222222_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.115
p=0.4788
p=0.1775
p=0.1865
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOMER3 | 222222_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1097
p=0.5607
p=0.268
p=0.3607
SmallAirway2
LargeAirway2
Trachea
Macrophage2

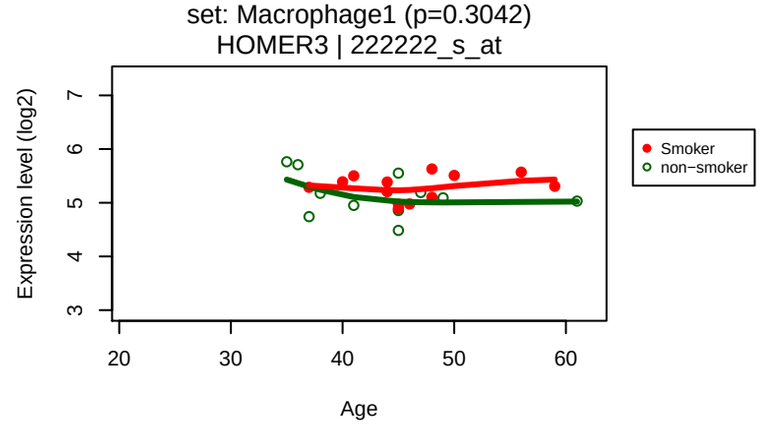

set: SmallAirway1 (p=0.2554)
HOMER3 | 222222_s_at
Expression level (log2)
Age
Smoker
non−smoker

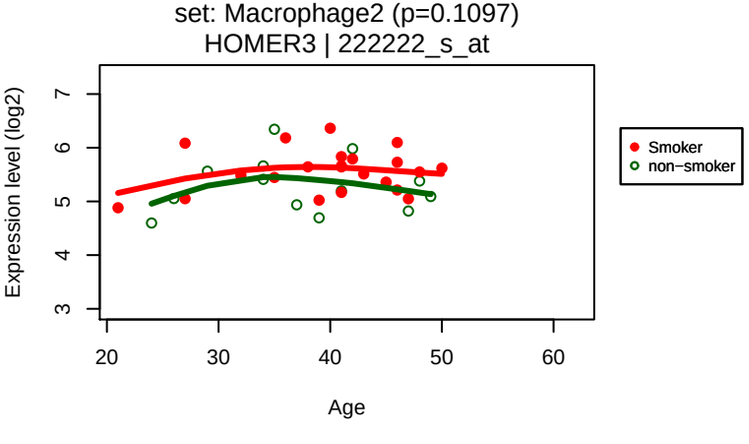

set: SmallAirway2 (p=0.6157)
HOMER3 | 222222_s_at
Expression level (log2)
Age
Smoker
non−smoker

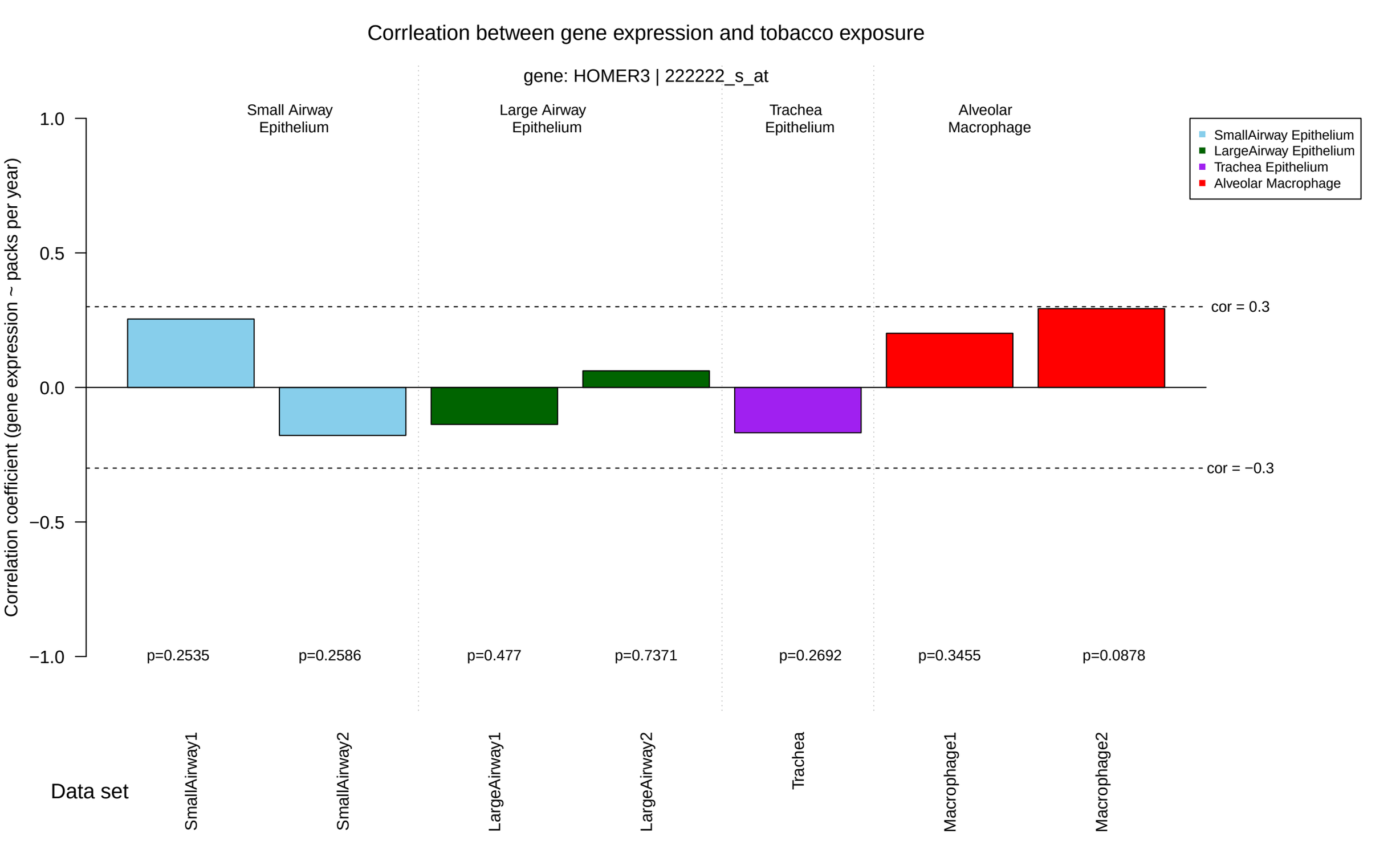

set: LargeAirway1 (p=0.8829)
HOMER3 | 222222_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2657)
HOMER3 | 222222_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0692)
HOMER3 | 222222_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3042)
HOMER3 | 222222_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1097)
HOMER3 | 222222_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOMER3 | 222222_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2535
p=0.2586
p=0.477
p=0.7371
p=0.2692
p=0.3455
p=0.0878
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2